FOR PRINCIPAL FITTING-TYPE IDEALS THE BETA-INVARIANT IS INDEPENDENT OF THE CHOSEN REPRESENTATIVE COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author's first sentence reads 'Observe that the fact that I_i(M^bullet, f) is a principal ideal implies that I_i(M^bullet, f) is a principal ideal by Lemma ...', stating the same ideal twice; that is the author's text and is reproduced unchanged. The author closes with 'This is a computation that we omit.', also kept verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The ideals I_i and J_i attached to a bounded complex of finite free modules and a nonzerodivisor, and their universal property. used throughout the counted proof span. Fixes the two ideals whose equality is asserted and the property used to reduce to summands.
- Well-definedness of I_i up to powers of f under change of representative complex. invoked at proof lines 28175-28178. Makes the principality hypothesis meaningful and supplies the reduction to a direct summand.
- Compatibility of the eta transform with direct sums. used at proof lines 28183-28189. Is the step that splits the displayed map into the contributions of M^bullet and Q^bullet.

Source reference. Original lemma 27893: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27893–27893. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a ring and let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ and $N^\bullet$ be two bounded complexes of finite free A-modules representing the same object in $D(A)$. Assume $I_i(M^\bullet, f)$ is a principal ideal for all $i \in \mathbf{Z}$. Then $J_i(M^\bullet, f) = J_i(N^\bullet, f)$ as ideals in A/fA.*

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Proof. Observe that the fact that $I_i(M^\bullet, f)$ is a principal ideal implies that $I_i(M^\bullet, f)$ is a principal ideal by Lemma 27893 and hence the statement makes sense. As in the proof of Lemma 27893 we may assume $N^\bullet = M^\bullet \oplus Q^\bullet$ for some trivial complex $Q^\bullet$, i.e.,

$$Q^\bullet = \ldots \to 0 \to A \xrightarrow{1} A \to 0 \to \ldots$$

where A is placed in degree j and $j+1$. Since η_f is compatible with direct sums, we see that the map

$$(1, d^i) : (\eta_f N)^i / f(\eta_f N)^i \longrightarrow f^i N^i / f^{i+1} N^i \oplus f^{i+1} N^{i+1} / f^{i+2} N^{i+1}$$

is the direct sum of the corresponding map for $M^\bullet$ and for $Q^\bullet$. By the universal property defining the ideals in question, we conclude that $J_i(N^\bullet, f) = J_i(M^\bullet, f) + J_i(Q^\bullet, f)$. Hence it suffices to show that $J_i(Q^\bullet, f) = 0$ for all i. This is a computation that we omit. $\square$